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END TERM EXAMINATION
SECOND SEMESTER [B.TECH] MAY-JUNE 2026

Paper Code: BS-112

Subject: Applied Mathematics-II

Time: 3 Hours

Maximum Marks: 60

Note: Attempt all questions as directed. Internal choice is indicated.

Q. No.	Question	Marks
Q1	Attempt any four of the following questions:	(4×5=20)
(a)	Determine the analytic function whose real part is $u = e^{2x}(x \cos 2y - y \sin 2y)$.	(5)
(b)	Evaluate $\int_0^{3+i} z^2 dz$ along the parabola $y^2 = x/3$.	(5)
(c)	State and prove the Second Shifting property of Laplace Transformation.	(5)
(d)	Show that the function $f(z) = z z $ is not analytic anywhere.	(5)
(e)	Find the linear fractional transformation (Möbius transformation) which maps the points $z = 0, 1, \infty$ to the points $w = -1, -i, 1$ respectively.	(5)
(f)	Obtain the Laplace transform of $f(t) = t^2 e^t \sin t$.	(5)
(g)	Find the Fourier cosine transform of $f(x) = e^{-x} + e^{-2x}$, $x > 0$.	(5)
(h)	Form a Partial Differential Equation from $z = f(y/x) + g(xy)$.	(5)
Q2 (a)	Determine the poles of the function $f(z) = z^3 / [(z-1)(z-2)^2]$ and the residue at each pole.	(5)
(b)	Prove that the function $f(z) = [x^3(1+i) - y^3(1-i)] / (x^2 + y^2)$, $z \neq 0$, and $f(0) = 0$ is continuous and Cauchy-Riemann equations are satisfied at the origin, yet $f'(0)$ does not exist.	(5)
OR		
Q3 (a)	Evaluate $\int_C [dz / ((z-1)(z-2)(z-3))]$ where C is the circle $ z = 4$.	(5)
(b)	Find the Laurent series of $f(z) = (7z-2) / (z^3 - z^2 - 2z)$ in the regions: (i) $0 < z+1 < 1$ (ii) $1 < z+1 < 3$.	(5)
Q4 (a)	Obtain the condition under which the mapping $w = (az+b) / (cz+d)$ maps a straight line of z-plane into a unit circle of w-plane.	(5)
(b)	Using residue theorem, evaluate $\int_C [e^z dz / ((z+1)^2(z-2))]$ where C: $ z-1 = 3$.	(5)
OR		
Q5 (a)	An electrostatic field in the xy-plane is given by the potential function $\phi = 3x^2y - y^3$. Find the Stream function $\psi(x, y)$.	(5)
(b)	Evaluate $\int_0^\pi [a d\theta / (a^2 + \sin^2\theta)]$, $(0 < a < 1)$.	(5)
Q6 (a)	Find the Laplace transform of: (i) $[(\cos at - \cos bt) / t]$ (ii) Find Inverse Laplace transform of $(s+2) / (s^2 - 4s + 13)$.	(5)
(b)	Solve, using Laplace Transform technique, the differential equation: $d^2y/dt^2 + 2(dy/dt) + 5y = e^{-t} \sin t$, where $y(0) = 0$, $y'(0) = 1$.	(5)
OR		
Q7 (a)	Obtain the Fourier series for the function $f(x)$ given by: $f(x) = 1 + (2x/\pi)$ for $-\pi \leq x \leq 0$, $f(x) = 1 - (2x/\pi)$ for $0 \leq x \leq \pi$.	(5)
(b)	Using Fourier transform to solve the equation $\partial\theta/\partial t = \partial^2\theta/\partial x^2$, $x > 0$, $t > 0$, subject to the conditions: (i) $u(0, t) = 0$, $t > 0$ (ii) $u(x, 0) = 1$ for $0 < x < 1$, and 0 for $x \geq 1$ (iii) $u(x, t)$ is bounded.	(5)
Q8 (a)	Using the method of separation of variables, solve the equation $\partial u/\partial x = 2(\partial u/\partial t) + u$, where $u(x, 0) = 6e^{-3x}$.	(5)

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Q. No.	Question	Marks
(b)	A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially in a position given by $y = a \sin(\pi x / l)$. If it is released from rest from this position, show that the displacement of any point at a distance x from one end at time t is given by: $y(x, t) = a \sin(\pi x / l) \cos(\pi c t / l)$.	(5)
	OR	
Q9 (a)	A rod of length l with insulated sides is initially at a uniform temperature u_0 . Its ends are suddenly cooled to 0°C and are kept at that temperature. Find the temperature function $u(x, t)$.	(5)
(b)	Solve the equation $\partial u / \partial t = \partial^2 u / \partial x^2$ with boundary conditions: $u(x, 0) = 3 \sin(n\pi x)$, $u(0, t) = 0$, and $u(l, t) = 0$, where $0 < x < l$, $t > 0$.	(5)

**** End of Question Paper ****